

Systems of Connected Particles: An Introduction

1. A flatbed truck of mass 1250 kg accelerates from a standstill. The driving force is 3000 N and the resistance force is 550 N.

(a) Find the acceleration of the truck.

Later, the same truck is loaded with a parcel of mass 200 kg on the flatbed. The truck again accelerates from a standstill, with the same driving and resistance forces acting on it.

- (b) By considering the whole system, find the acceleration of the truck.
 - (c) Draw a diagram showing the forces acting on the parcel, indicating which body is exerting each of the forces. Find their magnitudes.
 - (d) Draw a diagram showing the forces acting on the truck. There shouldn't be any unknowns – you can check that Newton's Second Law holds when applied to the truck.
2. My car has mass 900 kg, while my mass is 70 kg. I'm driving along and apply the brakes, generating a braking force of 1600 N. There are other resistance forces acting on the car, totalling 350 N.
 - (a) What is the deceleration of the car?
 - (b) What forces does the car exert on me?
3. A man of mass 70 kg stands in a lift of mass 750 kg. The lift cable exerts an upwards force of 8500 N on the lift.
 - (a) Find the acceleration of the lift.
 - (b) Find the force of the lift on the man.
 - (c) Draw a force diagram for the *lift*. There shouldn't be any unknowns – you can check that Newton's Second Law holds.